CS1632:
Software Verification

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Software Verification (a.k.a. Formal Verification)

- Proving one or the other about a program:
  - Program has no defect
  - Program has defects (and find all of them)
- What!?



… with some caveats ☺

Methods of Formal Verification

- Theorem Proving
  - Deducing postcondition from precondition through math

Caveat

- Model Checking
  - Given a finite state model of a system, exhaustively checking all the states to see if model meets a given specification

Caveat

Theorem Proving

Deducing postconditions from preconditions through math

Hoare Logic Theorem Proving

- Hoare Logic: Proves a Hoare Triplet through mathematical deduction
- Hoare Triplet: {Precondition} Program {Postcondition}
  - Meaning: Given Precondition and Program, Postcondition is always true

- Examples of Hoare Triplets (x is a variable, A and B are constants):
  - { True } x = A { x == A }
  - { x == A } x = x + B { x == A + B }
  - { x == A } if (x < 0) then x = -x { x == |A| }
  - { True } if (x < 0) then x = -x { x >= 0 }

Proof is done by composing Hoare Logic Triplets

- Suppose we wanted to prove the following assertion passes:
  x = -5;
  x = x + 3;
  if (x < 0) then x = -x;
  assertEquals(2, x); // prove this passes
- Composition of Hoare Logic Triplets:
  1. { True } x = -5 { x == -5 } and { x == -5 } x = x + 3 { x == -5 + 3 == -2 }
     → { True } x = -5; x = x + 3 { x == -2 }
  2. { True } x = -5; x = x + 3 { x == -2 } and { x == -2 } if (x < 0) then x = -x { x == |-2| == 2 }
     → { True } x = -5; x = x + 3; if (x < 0) then x = -x { x == 2 } → `Proof Complete!`

- Proof can be generated by human or automated theorem prover

Theorem Proving Advantages

- Can prove large programs with many (infinite) states
  - In contrast, model checker needs to visit each state to verify property is true
  - E.g. to prove { True } if (x < 0) then x = -x { x >= 0 }
    → Model checker needs to verify postconditions by visiting all values of x

- Leads programmer to a deeper understanding of the program
  - After spending weeks proving the program is correct, a natural outcome
  - But really, it does lead to some fundamental insights about your program

- Automated theorem provers exist (nowadays with help of AI)
  - Automated proof checkers also exist to catch AI (and human) hallucinations

Theorem Proving Disadvantages

- Source code is often not easily translated to math
  - C and Java code with side-effects are not easily translatable to math (Compared to functional languages like Haskell with pure functions)
  - There is no mathematical equivalent to complex data structures like graphs

- Automated provers are difficult to use in practice
  - Even if proof is automated, program specifications need to be hand-written
    - Often writing mathematical specifications for code takes as much effort as coding itself
  - Automated proofs are often too long or complex for human understanding
    - In one report by Motorola, an automated proof was 25 MB long (more than 100 pages)
    - Hard for humans to check that the proof didn't exploit a loophole in the specifications

Industry Reception

- Used only in niche markets where correctness is paramount
  - Mission critical systems, cryptography libraries, OS kernels
  - Proof for seL4 OS microkernel: https://github.com/seL4/l4v
    - 9,000 lines of C code resulted in 200,000 lines of Hoare Logic proof checked by Isabelle
    - Had to jump through hoops like translating C code to Haskell for easy verification
      1. Prove Haskell code meets OS specifications (confidentiality, integrity, availability, …)
      2. Prove C code (carefully written to minimize side-effects) is functionally equivalent to Haskell code
- The rest of the industry would like a “push button” solution
  - That is something that Model Checking can provide

Model Checking

Given a finite state model of a system, exhaustively checking whether this model meets a given specification

The Model Checking Problem

- Does the system model satisfy system properties?
- System model can be either:
  - The source code itself
  - Or some abstract model derived from source code
- System properties are specifications of system behavior
  - a.k.a. invariants, as they must invariably hold for every state

Examples of System Properties

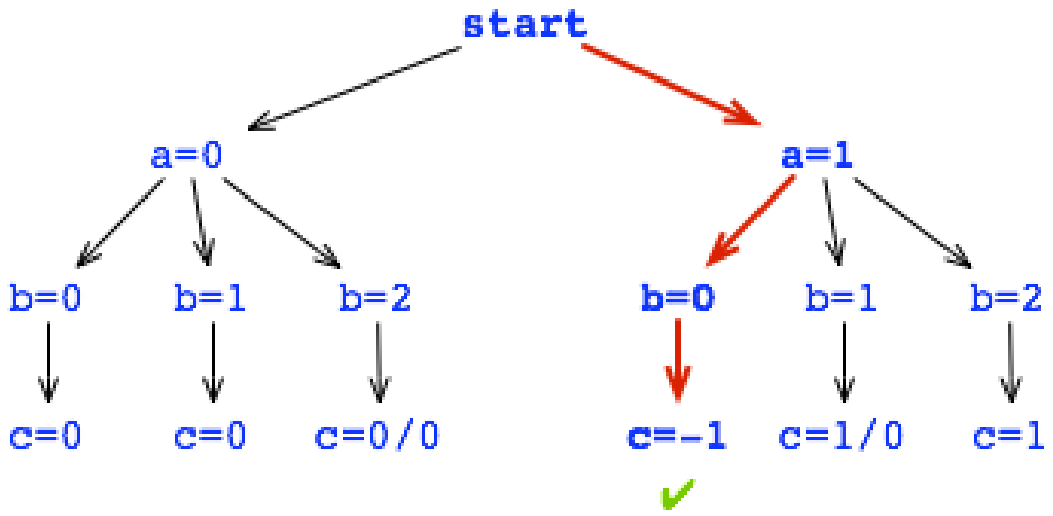
- Memory related properties
  - No out of bounds array accesses
  - No null references
  - No leaks, double-free, access after free (in C/C++)
- Thread related properties
  - No dataraces when threads access shared data
  - No deadlocks when using locks to synchronize threads
- User defined properties
  - Embedded as assertions in source code or part of property-based test

Conventional Testing

Given this code:

```
int a = random.nextInt(2);
int b = random.nextInt(3);
int c = a/(b+a -2);
```

Testing is unrepeatable
- It has 6 different outcomes!



① Random random = new Random()

② int a = random.nextInt(2)

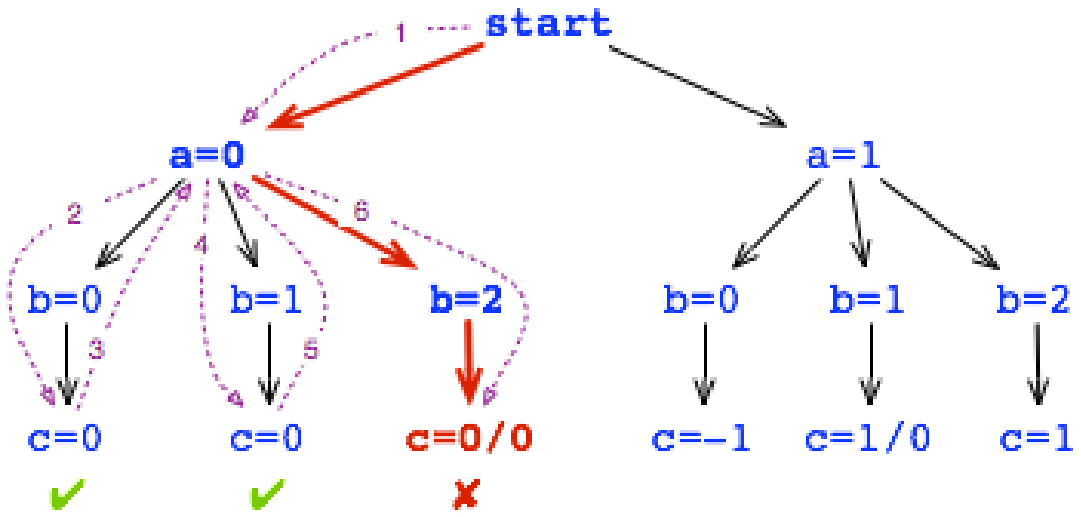
③ int b = random.nextInt(3)

④ int c = a/(b+a -2)

Model Checking

Given this code:

```
int a = random.nextInt(2);
int b = random.nextInt(3);
int c = a/(b+a -2);
```



Bug is always found!
(through exhaustive searching)
If none found, guaranteed correct!

① `Random random = new Random()`

② `int a = random.nextInt(2)`

③ `int b = random.nextInt(3)`

④ `int c = a/(b+a -2)`

Java Path Finder (JPF)

A Model Checker for Java

Java Path Finder (JPF)

- A model checker for Java: Uses all the principles we learned
  - Called Path Finder because it explores all paths in finite state machine
  - Concrete model checker: system model is the Java Virtual Machine
    → No need to translate source code into abstract model

- Developed and maintained by NASA
  - To model check code for their space missions
  - Open Source / Apache License Version 2
  - Released 2010, still actively maintained and extended

JPF VM enumerates all random states

```
int a = random.nextInt(2);
int b = random.nextInt(3);
int c = a/(b+a -2);
```

Not shown, but also generates a "trace" of random values chosen

```
-bash-4.2$ ./runJPF.sh Rand.jpf
JavaPathfinder core system v8.0 (C) 2005-2014 United States Government.
…
====================================================== error 1
gov.nasa.jpf.vm.NoUncaughtExceptionsProperty
java.lang.ArithmeticException: division by zero
        at Rand.main(Rand.java:34)
…
```

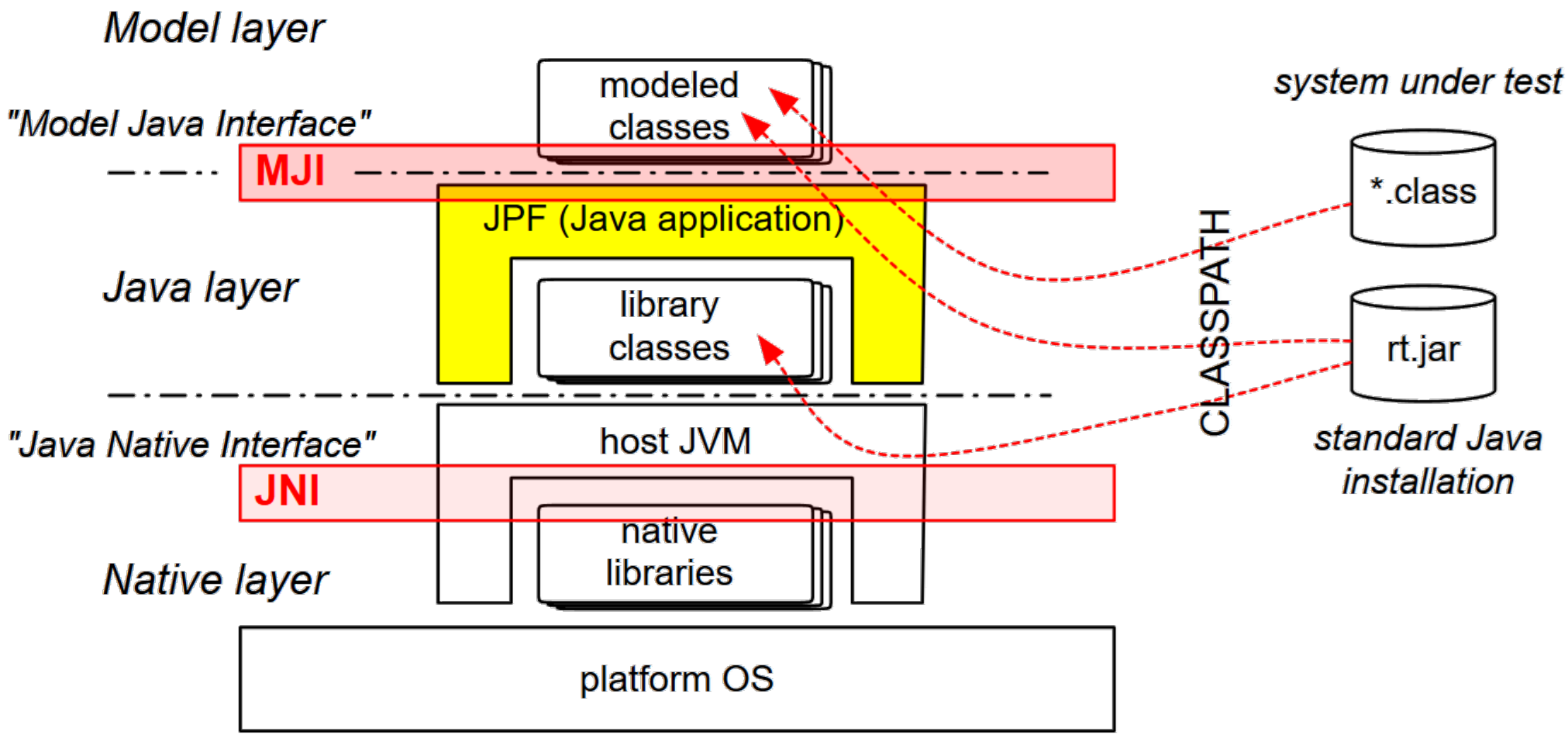
Where in code exception happened

JPF VM enumerates all thread interleavings

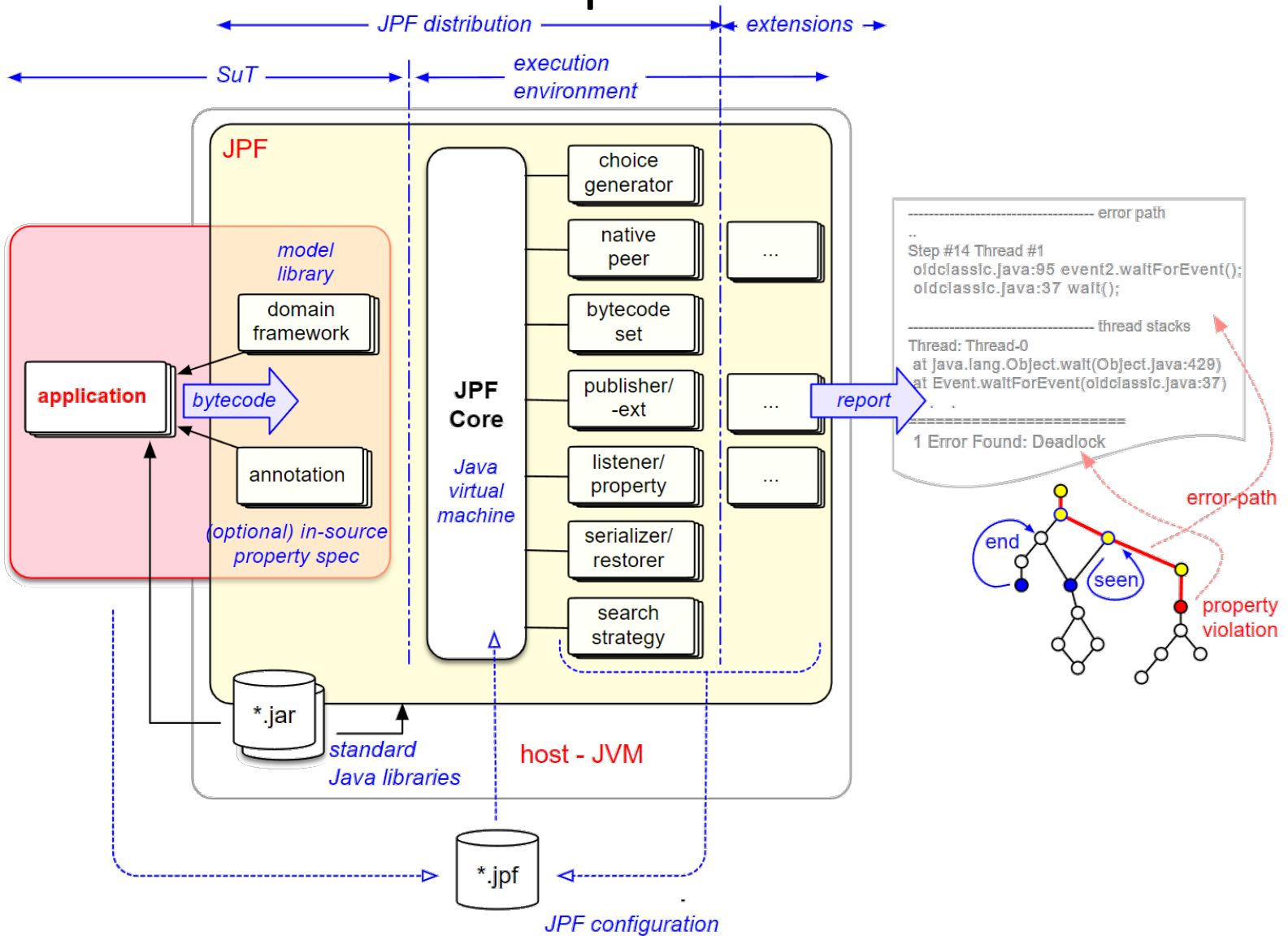
```
-bash-4.2$ ./runJPF.sh Interleaving.jpf
JavaPathfinder core system v8.0 (C) 2005-2014 United States
Government.
…
========================================== search started
main main Thread-1 Thread-1
main Thread-1 main Thread-1
main Thread-1 Thread-1 main
Thread-1 main main Thread-1
Thread-1 main Thread-1 main
Thread-1 Thread-1 main main
```

Able to explore all interleavings and generate all possible outputs!

App executes on top of Java Path Finder JVM



Java Path Finder is a special Java Virtual Machine



Java Path Finder Usage

Tips on how to use JPF

Java Path Finder: Example Configuration File

```
# Target class main method to run.  In this case, we are invoking the main method of Rand.
target = edu.pitt.cs.Rand
target.args =

# Don't stop at the first error.  Report all errors.
search.multiple_errors = true

# Properties checked on every state.  Only uncaught exceptions are enabled here.
search.properties = gov.nasa.jpf.vm.NoUncaughtExceptionsProperty

# On property violation, print the error, the instruction trace in the path to the violation
report.console.property_violation=error,trace
```

Java Path Finder: Example Report

JavaPathfinder core system v8.0 - (C) 2005-2014 United States Government. All rights reserved.

========== system under test

edu.pitt.cs.Rand.main() → **Main method you are testing**

========== search started: 3/27/20 12:38 AM

PROGRAM OUTPUT → **Output of your program if any**

========== results

no errors detected → **Errors will be listed if there are exceptions or assertion failures**

========== statistics

elapsed time: 00:00:06 → **Time elapsed for testing in hours:mins:seconds**

states: new=4155,visited=3529,backtracked=7684,end=467 → **States created while model checking**

…

========== search finished: 3/27/20 12:38 AM

Verify API: Enumerating Input Values

- Suppose you want to prove the following main method correct:

```
public static void main(String[] args) {
   int x = Integer.parseInt(args[0]);
   int y = Integer.parseInt(args[1]);
   int diff = x - y;
   if (x > y) assert diff > 0;
   if (x < y) assert diff < 0;
   System.out.println(x + " - " + y + " = " + diff);
   return;
}
```

- And you want to prove it correct for a set of command line arguments

Verify API: Enumerating Input Values

- If you wanted to verify `x = 3,4,5` in combination with `y = 4,6`:

```
public static void main(String[] args) {
   int x = Verify.getInt(3, 5);
   int y = Verify.getIntFromList(4, 6);
   int diff = x - y;
   if (x > y) assert diff > 0;
   if (x < y) assert diff < 0;
   System.out.println(x + " - " + y + " = " + diff);
   return;
}
```

- Verify API: verifies program for specified set of input values
  - Creates a path for each value in the set in the state machine, and travels down each path
  - Functionally, `Verify.getInt(3,5)` acts similarly as `(new Random()).getInt(3,6)`

Verify API: Java Path Finder Report

JavaPathfinder core system v8.0 - (C) 2005-2014 United States Government. All rights reserved.

========== system under test

JPFTester.main()

========== search started: 3/27/20 12:38 AM

3 - 4 = -1

3 - 6 = -3

4 - 4 = 0

4 - 6 = -2

5 - 4 = 1

5 - 6 = -1

Output of:

```
x = Verify.getInt(3, 5)
y = Verify.getIntFromList(4, 6)
```

========== results

no errors detected

…

========== search finished: 3/27/20 12:38 AM

State Space Explosion

When it happens and how to deal with it

State Space Explosion Problem

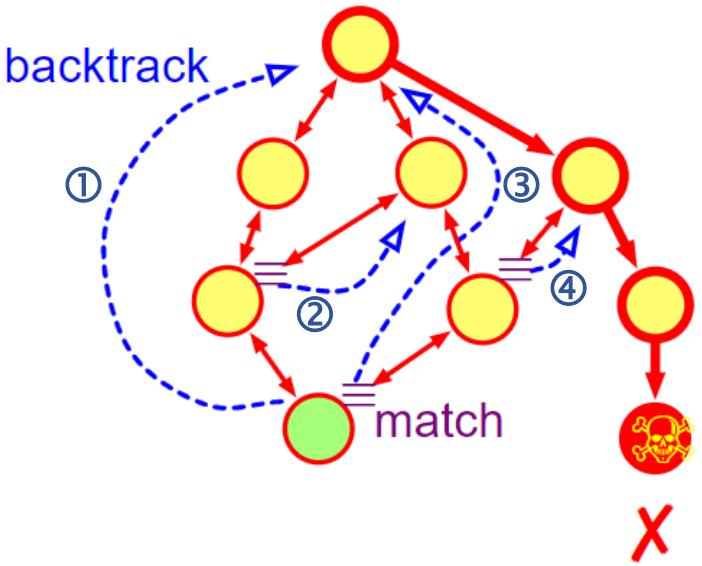
- Some programs may have (close to) an infinite states in their state machines
  - May run into memory limitations (can't contain entire state graph)
  - May run into time limitations (can't explore entire graph within allotted time)

  → This is called the **State Space Explosion Problem**

- Single reason preventing wide adoption of model checking

- **State matching & backtracking** alleviates the problem (but not all)

State Exploration with Matching & Backtracking



Circles: Program states
Arrows: State transitions

- State divergence happens when there is a *choice*
  - Value from random number generation or user input
  - Choice of thread to run among multiple threads
- **Match**
  - When next state matches a previously visited state
  - Backtrack to not repeat work
- **Backtrack**
  - On reaching terminal state or when there is a match
  - Go to closest previous state with unexplored transition

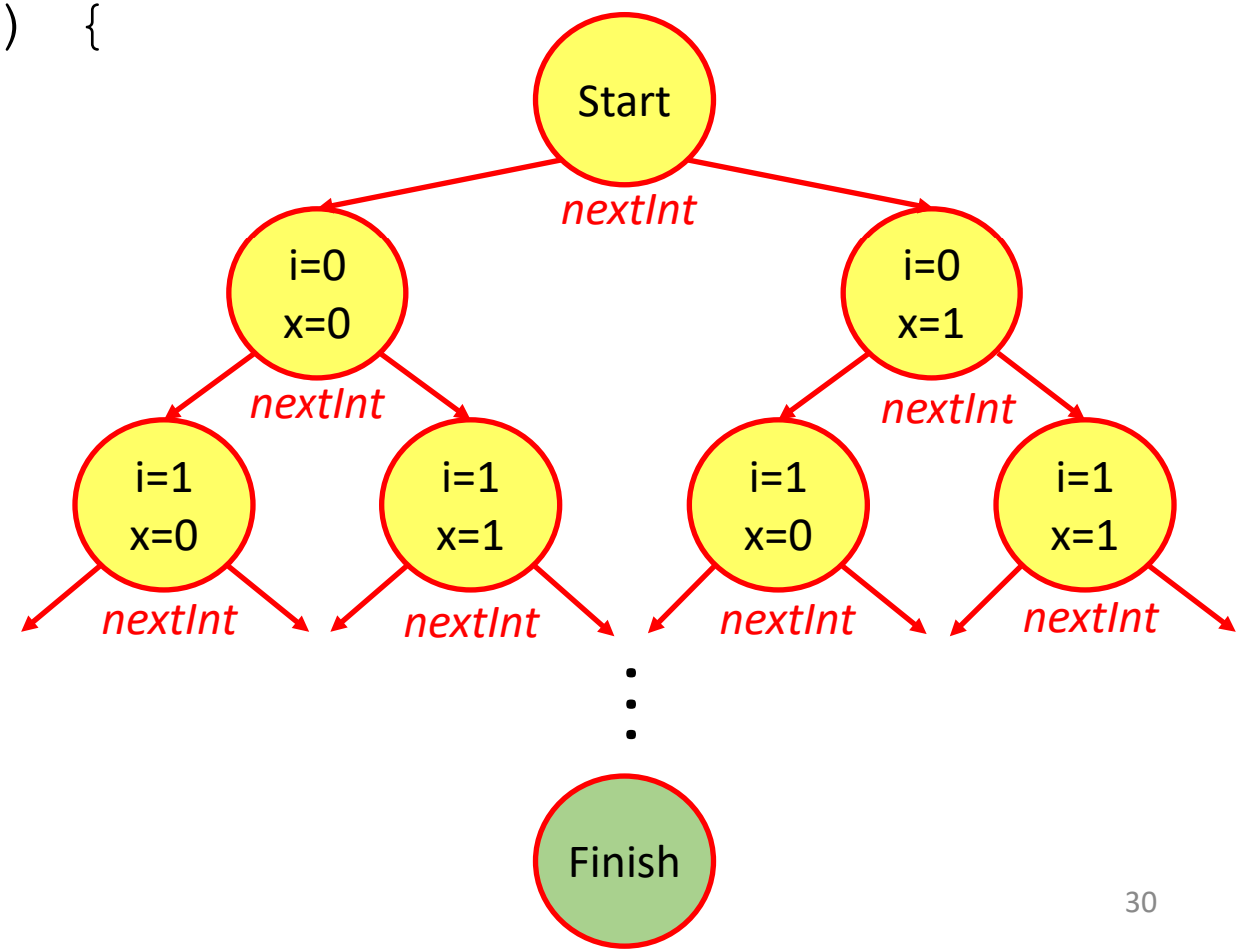
→ Ensures **each unique state is visited only once**

Matching & Backtracking: Example 1

- Given below code:

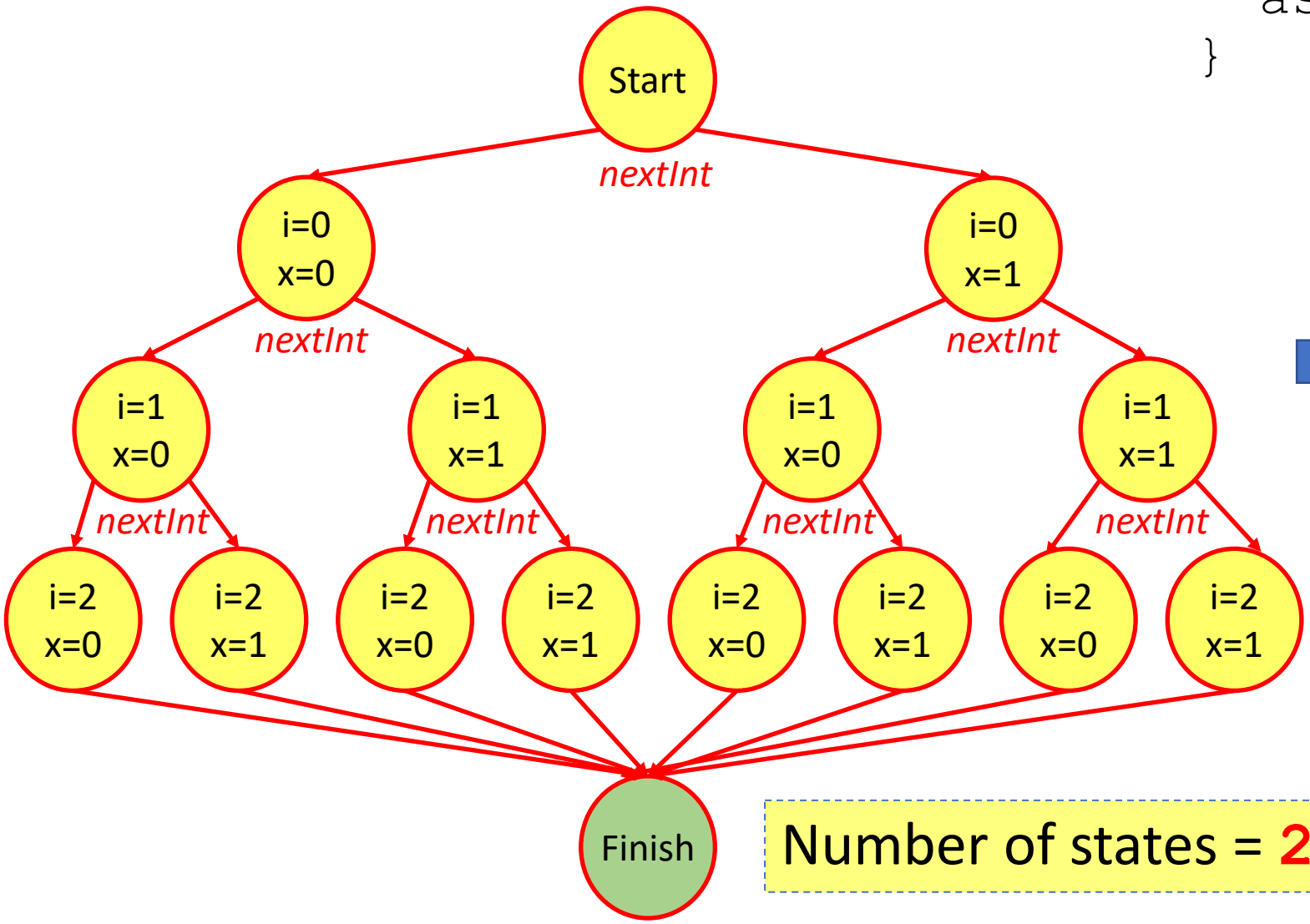
```
for(int i = 0; i < N; i++) {
  x = rand.nextInt(2);
  assert x < 2;
}
```

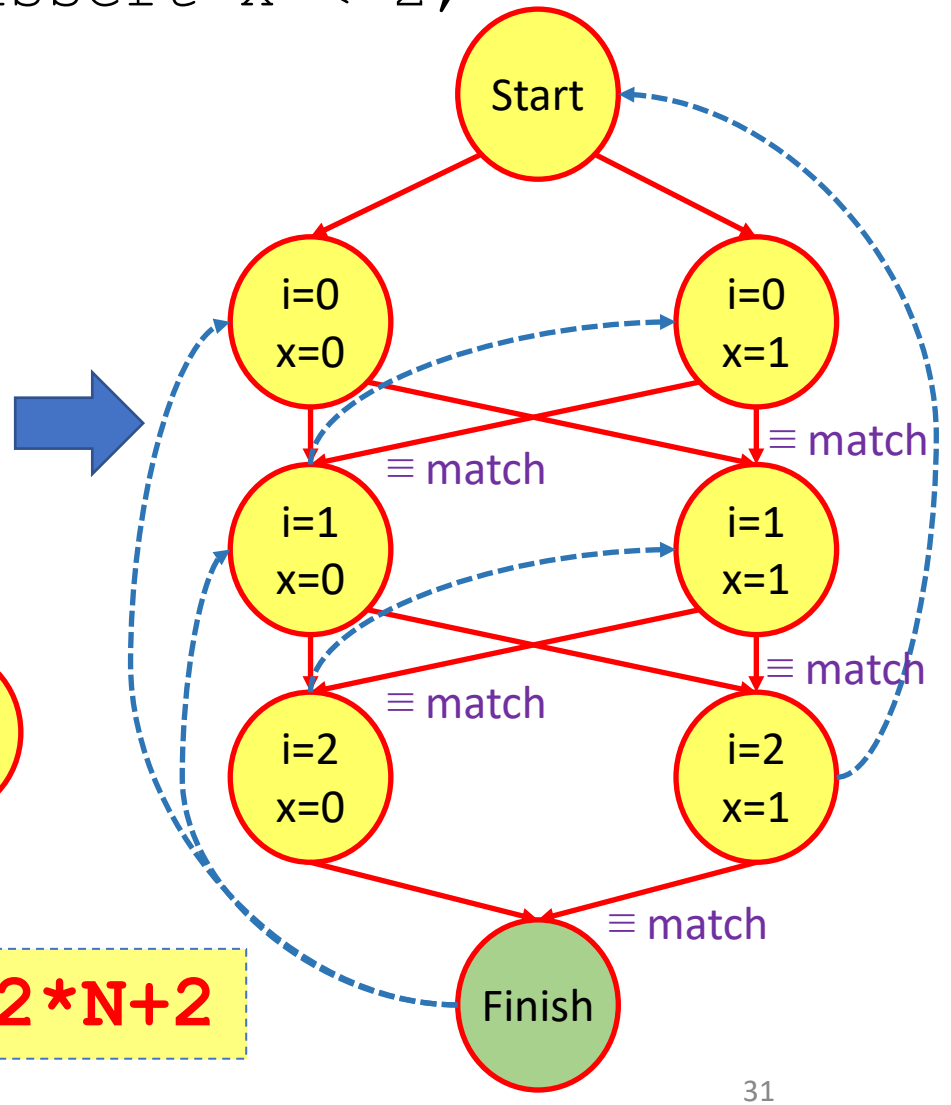
- Potential number of states = 2^{N+1}



Matching & Backtracking on

```
for(int i = 0; i < 3; i++) {
  x = rand.nextInt(2);
  assert x < 2;
}
```





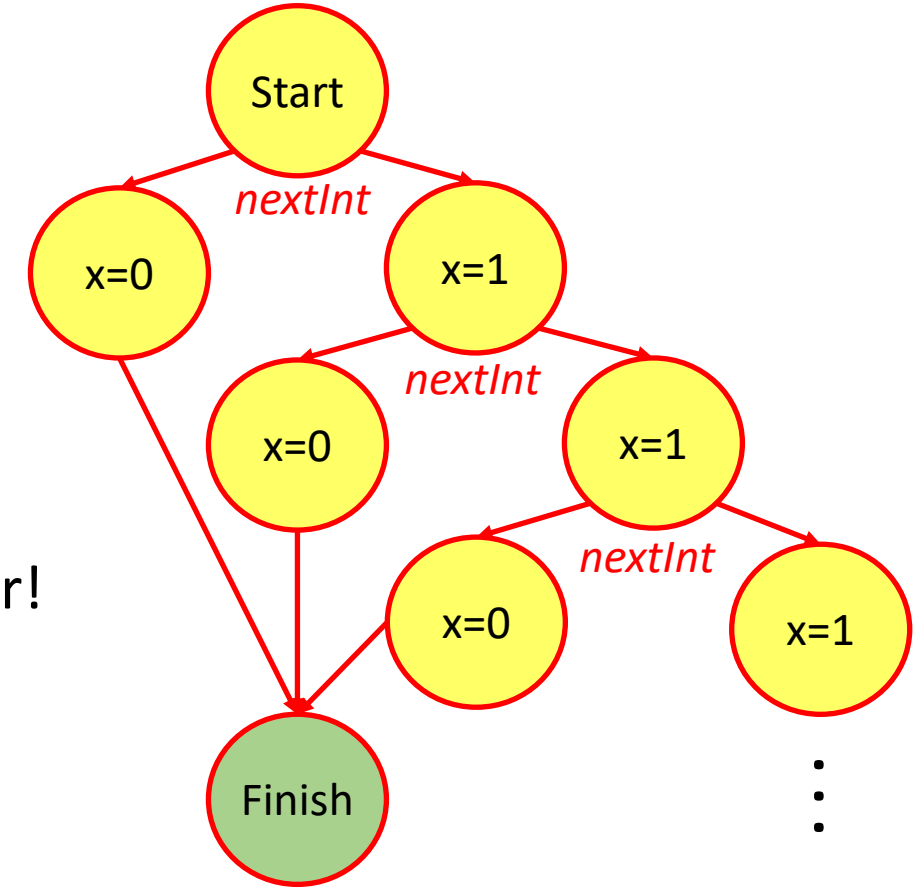
Number of states = **2*N+2**

Matching & Backtracking: Example 2

- Given below code:

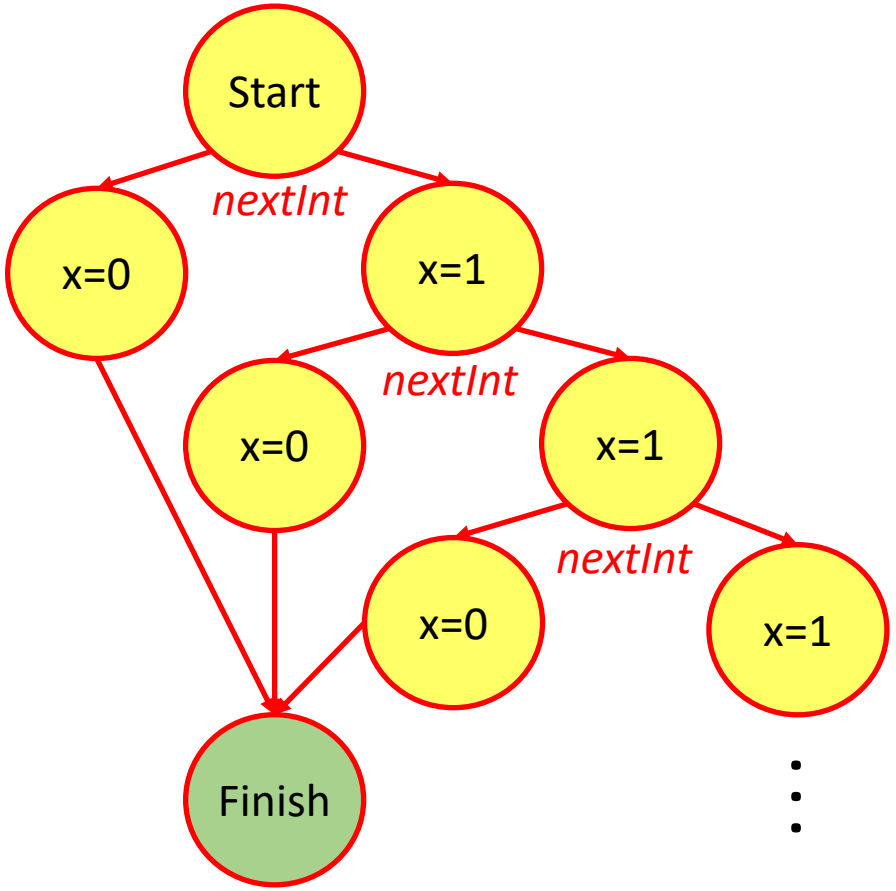
```
while(true) {
   x = rand.nextInt(2);
   if(x == 0) break;
   assert x < 2;
}
```

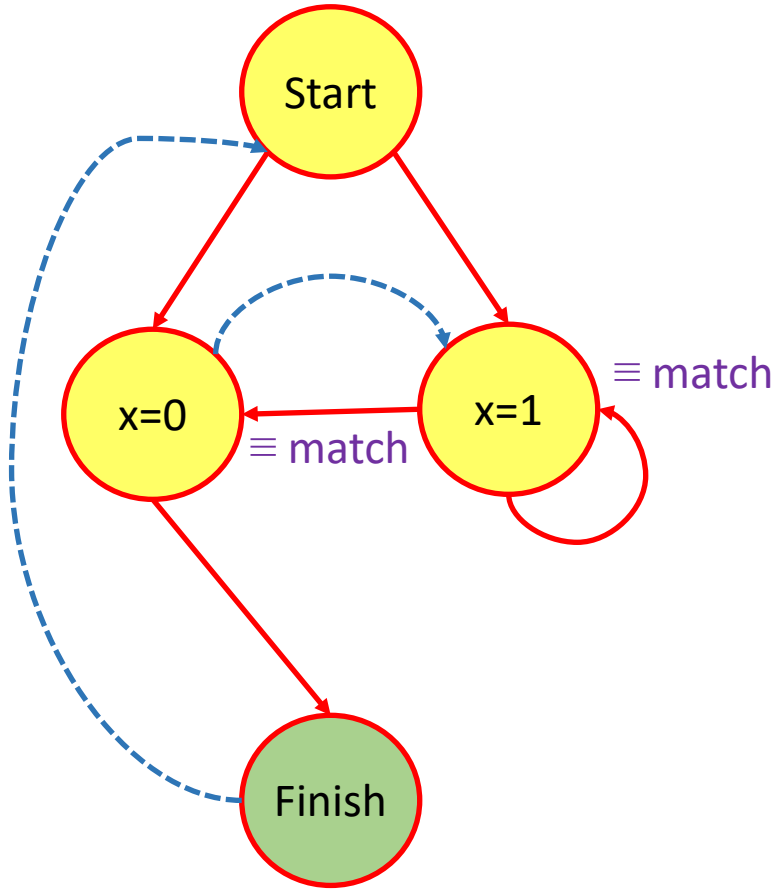
- Model checker can potentially go on forever!
  - Will keep creating states to the right



Match & Backtrack on

```
while(true) {
   x = rand.nextInt(2);
   if(x == 0) break;
   assert x < 2;
}
```





Number of states = **4**

State Exploration Pseudo-code (DFS example)

```
Hashtable states_seen;
Stack pending;

pending.push(initial_state);
while(!pending.empty()){
      current = pending.pop();
      if(current in states_seen)
            continue;  // match! Backtrack.
      check current for correctness;
      states_seen.insert(current);
      for transition T in current {
            successor = execute transition T on current;
            pending.push(successor);
      }
}
```

State Space Explosion Sometimes is Unavoidable

- Even with state matching, sometimes state space explodes
  - Due to variables that take on an unbounded set of values during execution

- Can filter out part of program state that causes explosion
  - E.g. `@FilterField` annotation on a field of a class filters out that variable
  - Filtered state is ignored for the purposes of state matching
    → Suppresses duplication of states that only differ by that filtered variable
    → May result in parts of full program state machine that are not traversed
  - State needing filtering typically not part of core functionality that is checked (e.g. event counters, statistics, logs, etc.)

Limiting State Creation Using @FilterField

```
public class WebServer {
  // For statistics gathering purposes
  int pageCounter;

  public void sendPage(String url) {
    pageCounter++;
    // Do the actual processing
  }
}
```

- `pageCounter` puts `WebServer` in a new state every time `sendPage` is called
  → Even when server is in the same state otherwise (e.g. servicing the same web page)
  → Leads to a lot of unnecessary state creation

Limiting State Creation Using @FilterField

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```

- `pageCounter` puts `WebServer` in a new state every time `sendPage` is called
  → Even when server is in the same state otherwise (e.g. servicing the same url)
  → Leads to a lot of unnecessary state creation
- `@FilterField` allows JPF to verify each url once, ignoring differences in `pageCounter`

JUnit Testing with Java Path Finder

JUnit is needed to check program-specific invariants

- The JPF can be natively configured to check many systems properties
  - Uncaught exceptions (a.k.a. crashes)
  - Infinite loops
  - Data races
  - Dead locks

  …

- JPF cannot check for program-specific properties it doesn’t know about
  - Testers can encode these invariant properties as assertions
  - Separation-of-concerns: separate out these assertions into a JUnit class

Create a new JPF virtual machine for each @Test

- The JUnit class is executed on the host Java virtual machine
  - All @Test, @Before, @After methods executed on the host JVM as usual

- For @Test methods that are JPF tests, create a new JPF VM for each
  - Why each? Why not just have one JPF VM execute all the tests?
  1. Each JPF VM may need a different setup (e.g. different properties verified)
  2. Also helps keeps the state space from compounding and exploding (e.g. second test starts a new traversal from each terminal state of first test)

- **`verifyNoPropertyViolation()`** API creates a new JPF VM
  - Call at the beginning of each JPF @Test method

`verifyNoPropertyViolation()` semantics

- Semantics of **`verifyNoPropertyViolation()`**:
  - Creates a new JPF virtual machine at the point of the call
  - Results in two virtual machines executing after the call
    - **Host** virtual machine: returns **false** on verifyNoPropertyViolation() call
      → Immediately return from the @Test method to move on to the next @Test to launch
    - **JPF** virtual machine: returns **true** on verifyNoPropertyViolation() call
      → Proceed to the body of the @Test to run it on top of the JPF VM

- Can take arguments that customizes JPF VM for that particular test
  - verifyNoPropertyViolation("+listener=gov.nasa.jpf.listener.PreciseRaceDetector")
    → Adds a race detector that listens during state traversal for data races

Pitfall: Expecting JPF VM to execute @Before

```
public class ArithmeticTest {
  private int x = 0;

  @Before public void setUp() {
    // Only executed on host virtual machine (JPF VM doesn't exist yet)
    // Host VM simply assigns x = 0 (doesn't know any better)
    x = Verify.getInt(-5, 5);
  }

  @Test public void testSquare() {
    if (verifyNoPropertyViolation() == false) {
      // This is the host virtual machine so return immediately.
      return;
    }
    // Bug! Test case is tested against x == 0 only.
    int y = x * x;
    assertTrue(y > 0);
  }
}
```

Pitfall: Fixed

```
public class ArithmeticTest {
  private int x = 0;

  public void setUp() {
    x = Verify.getInt(-5, 5);
  }
  @Test public void testSquare() {
    if (verifyNoPropertyViolation() == false) {
      // This is the host virtual machine so return immediately.
      return;
    }
    // Start by calling setUp() on the JPF virtual machine.
    setUp();
    int y = x * x;
    assertTrue(y > 0);
  }
}
```

Backup

Holy Grail of Formal Verification

- Soundness
  - If no defect is reported, then the program does not fail
  - No false negatives
- Preciseness
  - If a defect is reported, then the program does fail
  - No false positives
- Termination
  - The verification terminates

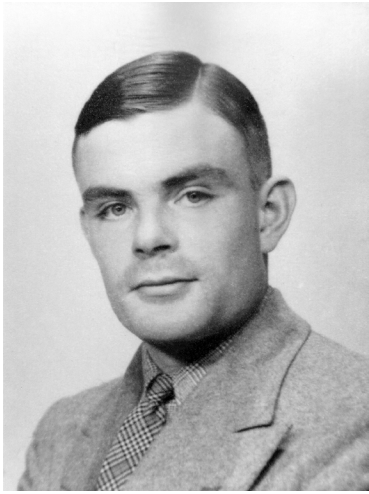
It is impossible to achieve the holy grail in general!

Formal Verification is Undecidable

$x \in$ Variable

$P \in$ Program = assert x | x++ | x-- |
$P_1 ; P_2$ | if x then P_1 else P_2 | **while x P**

- Assertion checking for even this simple language is undecidable!
- "The **Halting Problem** cannot be solved for *all* possible programs (for a Turing-complete language)" - *Alan Turing (1936)*
- Silver-lining: But for *some (most)* programs it can be solved



Concrete and Abstract Model Checking

- Concrete model checking
  - States in model are actual concrete program states
  - As in, your program stack and your program heap

- Abstract model checking
  - States in model are some abstraction of actual program states
  - Abstraction is done in hopes of reducing the state explosion problem
  - Typically tradeoffs accuracy for efficiency

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Abstract Model Checking

- Requires an intermediate description of abstract model
  - Describes the system at a high level
  - Throws away implementation details
- Good for checking designs, rather than implementations
  - Success stories: hardware circuits, cache-coherence protocols
- Problem: Specifying an abstract model is HARD for large systems
  - What you check is not what you run!
  - As the system evolves model has to be updated
  - Manual extraction of abstract model can miss or introduce errors

Automatically Extracting the Model

- Statically analyze the code to generate a model
  - Models usually mimic the implementation

Murphi abstract model

```
Rule "PI Local Get (Put)"
 1:Cache.State = Invalid
    & ! Cache.Wait
 2: & ! DH.Pending
 3: & ! DH.Dirty ==>
  Begin
 4: Assert !DH.Local;
 5: DH.Local := true;
 6: CC_Put(Home, Memory);
  EndRule;
```

Flash Memory Driver Implementation

```
void PILocalGet(void) {
     // ... Boilerplate setup
2    if (!hl.Pending) {
3      if (!hl.Dirty) {
4!      // ASSERT(hl.Local);
          ...
5       hl.Local = 1;
6       PI_SEND(F_DATA, F_FREE, F_SWAP,
                F_NOWAIT, F_DEC, 1);
```

Automatic Model Extraction

- Examples
  - FeaVer : C program -> Promela (SPIN) model
  - Bandera: Java -> Bandera model
- Features
  - Sophisticated property-driven slicing techniques
  - Can throw away state unrelated to property that is being proved
- Problems
  - Not all primitives are available in the modeling language
    - Pointers, dynamic object creation, dynamic threads, exceptions
  - A precise-enough slice could be as large as the program itself

Concrete and Abstract Model Checking

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Concrete Model Checking

- Code as the model – directly execute the code!

- Concrete model checkers
  - Verisoft (C/C++) – Bell Laboratories
  - CBMC (C/C++) – Oxford University
  - Java Path Finder (Java) – NASA

Language and Concrete Model Checking

- Not many programming languages have a dedicated model checker

- Fortunately, you can convert most languages to Java bytecode
  - JavaScript, Python, Ruby, Lua, …
  - Even (for a limited set of) C / C++
  - And then, you can model check the bytecode using a JVM

State Space Reduction Techniques

- State space can be infinite (or very large)
  - Try exploring as many behaviors as possible (likely you can’t explore all)
  - Focus on precision (finding defects accurately)

- Techniques to reduce memory consumption due to state space
  - State collapsing
  - Heuristic state approximation
  - Hash compaction
  - Heap canonicalization
  - Symbolic execution

State Space Reduction Techniques

- State collapsing
- Heuristic state approximation
- Hash compaction
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- Symbolic execution

State collapsing

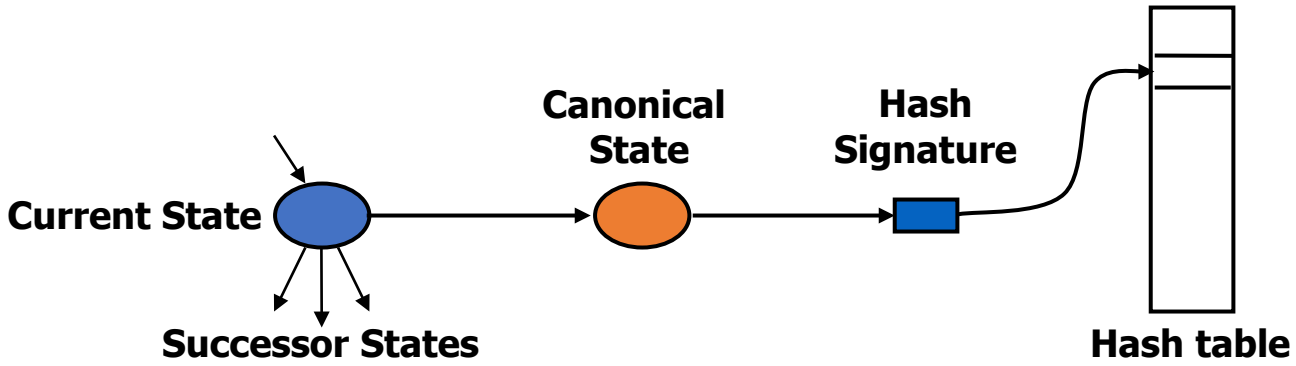
- Typically a state transition involves changing very small state
  - Updating a local variable in the stack
  - Creating a new object on the heap
- Instead of storing the entire state each time in hash table …
- Store the delta (change) from the previous state in hash table

State Space Reduction Techniques

- State collapsing
- Heuristic state approximation
- Hash compaction
- Heap canonicalization
- Symbolic execution

Heuristic state approximation

- Explore one out of a (large) set of equivalent states
- Canonicalize (unify) states before hashing



- Example: suppose an int value has two equivalence classes
  - When hashing to check for match with an already visited state, Unify all values in equivalence class to one chosen value

  → Leads to a drastic reduction in visited states

  → Can also lead to missed defects – **unsound**

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Hash compaction

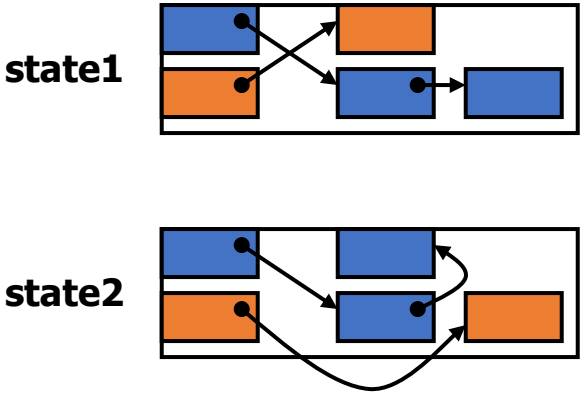
- Only store hash of state in the hash table
  - Do not store the actual state
  - A state match is determined solely by an equality check on the hash
- Might miss defects due to hash collisions – **unsound**
  - Two states that are different may be stored as the same hash
  - Means some states will not be visited as a result
- But orders of magnitude memory savings
  - Can compact 100 kilobyte state to 4-8 bytes!

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Heap canonicalization

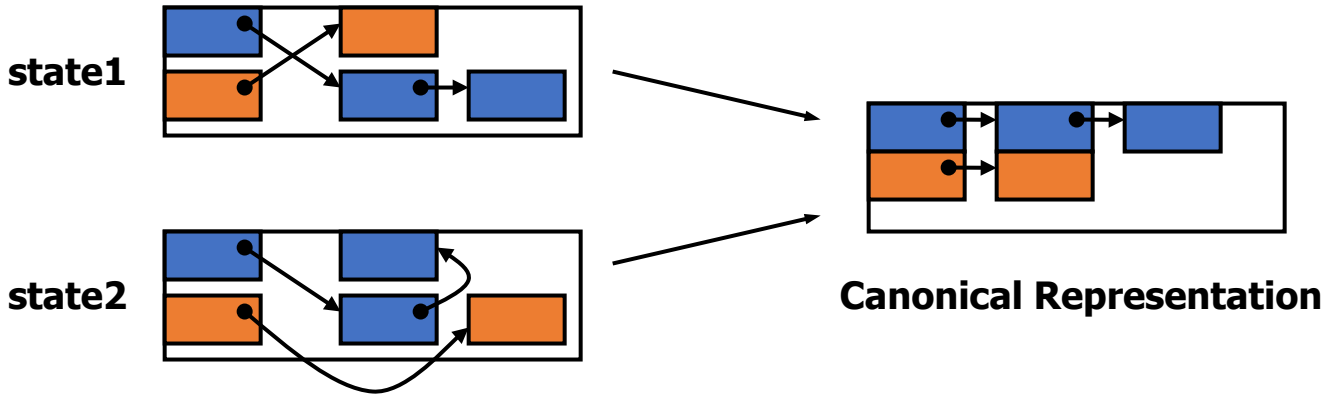
- Problem: two logically equivalent program states appear different because of differences in heap layout
- Example:



- Are the two states logically different?
  - No! But appears different due to different reference pointer values.

Heap canonicalization

- Solution: Canonicalize heap to unify layout
- Example:



- Canonical layout can be found by doing a fixed traversal of heap
  - DFS: Depth first search, or BFS: Breadth first search
- Note: can do it incrementally on each heap modification w/o full traversal

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To be continued …